

# Lecture 10 - Class Activity: Multiple Regression

Bill Perry

## Analysis of Net Primary Production in Forests: A Modern Tidyverse Approach

*Based on Michaletz et al. (2014) data*

### Introduction

This analysis examines the relationships between Net Primary Production (npp) and various climate and forest characteristics across global forest sites. We'll explore multicollinearity, model selection, and variable transformations.

### Key Learning Objectives:

- Understand multicollinearity in multiple regression
- Learn model diagnostics and assumption checking
- Practice variable selection techniques
- Apply data transformations appropriately

### Load Required Packages

```
# Load required packages
library(tidyverse) # For data manipulation and visualization
library(car)       # For regression diagnostics (VIF, etc.)
library(corrplot)  # For correlation plots
library(GGally)    # For pairs plots
library(broom)     # For tidy model outputs
library(performance) # For model performance metrics
library(see)       # For better diagnostic plots
```

### Load and Explore the Data

```
# Load the forest npp data
forest_df <- read_csv("data/michaletz_etal_2014_clean.csv")

# Display top lines
head(forest_df)
```

```
# A tibble: 6 × 8
  npp    age biomass season  temp precip    teb leaf
<dbl> <dbl>   <dbl>   <dbl> <dbl>   <dbl> <dbl> <chr>
1  2084  104.  18198.    11 25.3   1888   0.55 broadleaf
2  2234  333.  54523.    12 26.9   2348.   0.6  broadleaf
3  2714  213.  41358.    12 26.9   2348.   0.6  broadleaf
4  2828  114.  31557.    12 26.7   2784.   1.25 broadleaf
5  2882  113.  21417.    12 26.7   2784.   1.25 broadleaf
6   774   79  11188.     3 -3.53   408.   1.8  broadleaf
```

## Data Preparation

Following the original analysis, we'll focus on the key variables from a cleaned dataframe

### 1. Initial Exploration: Variable Relationships and Multicollinearity

#### Correlation Matrix and Visualization

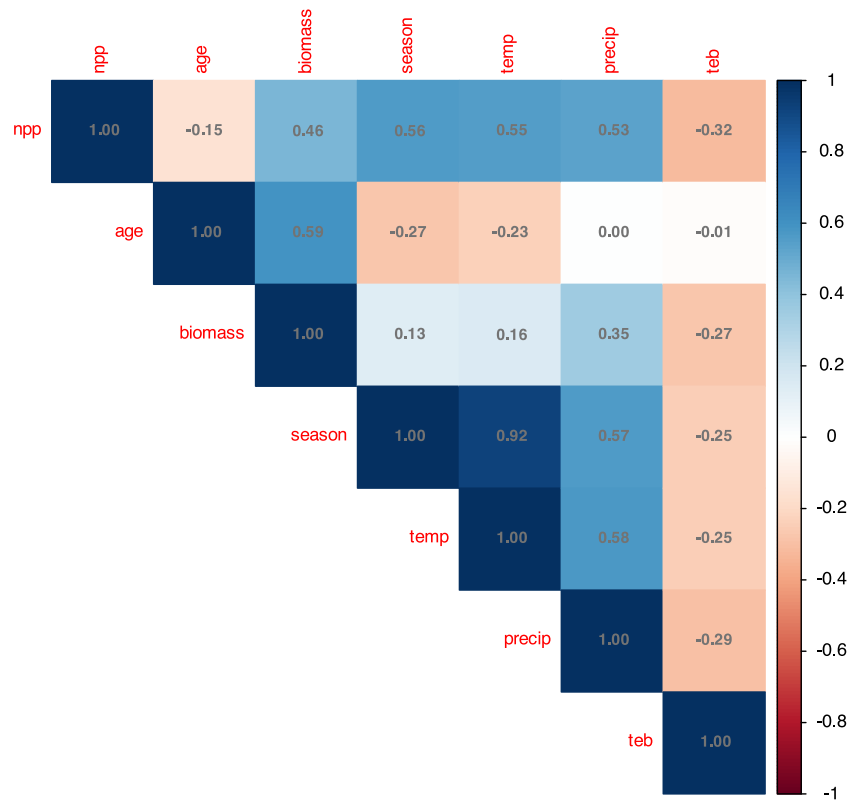
```
# Create correlation matrix for numeric variables only
num_vars <- forest_df %>%
  select_if(is.numeric)

# Calculate correlation matrix
cor_matrix <- cor(num_vars, use = "complete.obs")

# Display correlation matrix
cor_matrix
```

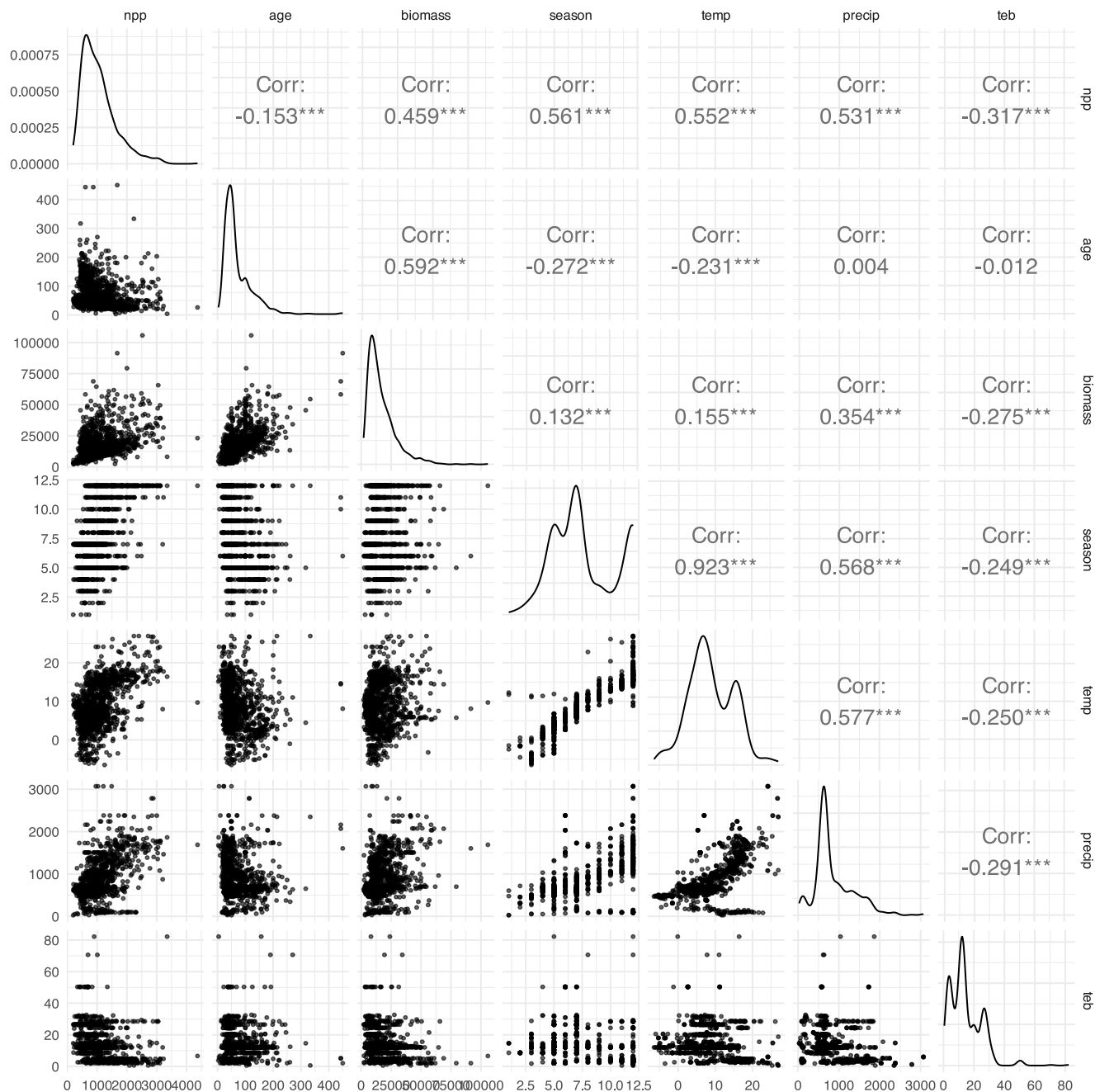
|         | npp         | age          | biomass    | season     | temp       | precip       |
|---------|-------------|--------------|------------|------------|------------|--------------|
| npp     | 1.0000000   | -0.153215979 | 0.4591083  | 0.5614272  | 0.5520590  | 0.531186333  |
| age     | -0.1532160  | 1.000000000  | 0.5923809  | -0.2715014 | -0.2307658 | 0.003642104  |
| biomass | 0.4591083   | 0.592380901  | 1.0000000  | 0.1317097  | 0.1551923  | 0.353901541  |
| season  | 0.5614272   | -0.271501387 | 0.1317097  | 1.0000000  | 0.9226905  | 0.567630275  |
| temp    | 0.5520590   | -0.230765849 | 0.1551923  | 0.9226905  | 1.0000000  | 0.576659967  |
| precip  | 0.5311863   | 0.003642104  | 0.3539015  | 0.5676303  | 0.5766600  | 1.000000000  |
| teb     | -0.3167187  | -0.011779155 | -0.2748724 | -0.2491779 | -0.2495479 | -0.291376661 |
| teb     |             |              |            |            |            |              |
| npp     | -0.31671865 |              |            |            |            |              |
| age     | -0.01177916 |              |            |            |            |              |
| biomass | -0.27487245 |              |            |            |            |              |
| season  | -0.24917787 |              |            |            |            |              |
| temp    | -0.24954794 |              |            |            |            |              |
| precip  | -0.29137666 |              |            |            |            |              |
| teb     | 1.00000000  |              |            |            |            |              |

```
# Create a visual correlation plot
corrplot(cor_matrix, method = "color", type = "upper",
  addCoef.col = "grey45", tl.cex = 0.8, number.cex = 0.7)
```



## Pairs Plot for Visual Inspection

```
# Create pairs plot to visualize relationships
# This replaces the original pairs() function with ggplot2
forest_df %>%
  select(-leaf) %>% # Exclude categorical variable for pairs plot
  ggpairs(
    upper = list(continuous = wrap("cor", size = 5)),
    lower = list(continuous = wrap("points", alpha = 0.6, size = 0.8))
  ) +
  theme_minimal()
```



## 2. Initial Multiple Regression Model

Let's start with a full model including all predictors:

```
# Fit initial model with all predictors (Model 1)
model_init <- lm(npp ~ age + biomass + season + temp +
  precip + teb + leaf, data = forest_df)

# Get model summary
summary(model_init)
```

Call:

```
lm(formula = npp ~ age + biomass + season + temp + precip + teb +
  leaf, data = forest_df)
```

Residuals:

| Min      | 1Q      | Median | 3Q     | Max     |
|----------|---------|--------|--------|---------|
| -1331.10 | -206.27 | -34.09 | 166.94 | 2760.41 |

Coefficients:

|             | Estimate   | Std. Error | t value | Pr(> t )     |
|-------------|------------|------------|---------|--------------|
| (Intercept) | 583.92959  | 53.81169   | 10.851  | < 2e-16 ***  |
| age         | -4.78822   | 0.28518    | -16.790 | < 2e-16 ***  |
| biomass     | 0.03154    | 0.00122    | 25.848  | < 2e-16 ***  |
| season      | 41.18220   | 9.49073    | 4.339   | 1.55e-05 *** |
| temp        | 4.61281    | 4.37372    | 1.055   | 0.291788     |
| precip      | 0.09674    | 0.02852    | 3.392   | 0.000716 *** |
| teb         | -2.12880   | 1.08408    | -1.964  | 0.049794 *   |
| leafneedle  | -267.00569 | 22.43078   | -11.904 | < 2e-16 ***  |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 351.7 on 1212 degrees of freedom

Multiple R-squared: 0.6415, Adjusted R-squared: 0.6394

F-statistic: 309.8 on 7 and 1212 DF, p-value: < 2.2e-16

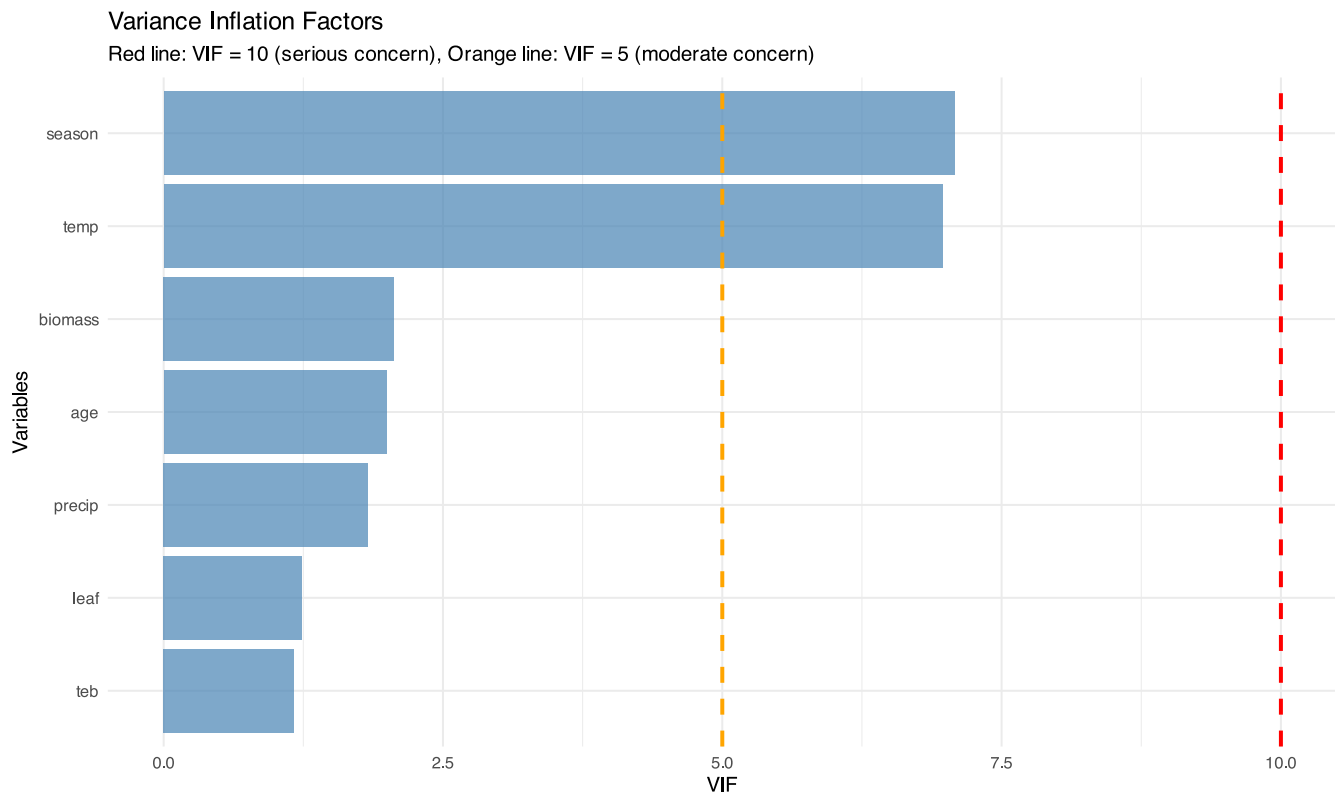
## Check for Multicollinearity

```
# Calculate Variance Inflation Factors (VIF)
vif_values <- vif(model_init)
vif_values
```

| age      | biomass  | season   | temp     | precip   | teb      | leaf     |
|----------|----------|----------|----------|----------|----------|----------|
| 1.993348 | 2.061766 | 7.079004 | 6.972147 | 1.831127 | 1.167007 | 1.236588 |

```
# Create a data frame for better visualization
vif_df <- data.frame(
  Variable = names(vif_values),
  VIF = as.numeric(vif_values)
) %>%
  arrange(desc(VIF))

# Visualize VIF values
ggplot(vif_df, aes(x = reorder(Variable, VIF), y = VIF)) +
  geom_col(fill = "steelblue", alpha = 0.7) +
  geom_hline(yintercept = 10, color = "red", linetype = "dashed",
    linewidth = 1) +
  geom_hline(yintercept = 5, color = "orange", linetype = "dashed",
    linewidth = 1) +
  coord_flip() +
  labs(
    title = "Variance Inflation Factors",
    subtitle = "Red line: VIF = 10 (serious concern), Orange line: VIF = 5 (moderate concern)",
    x = "Variables",
    y = "VIF"
  ) +
  theme_minimal()
```



### Address Multicollinearity by Removing Growing Season

Based on the original analysis, season and Temperature are highly correlated. Let's remove season:

```
# Model 2: Remove season due to multicollinearity
model_2 <- lm(npp ~ age + biomass + temp + precip + teb + leaf,
              data = forest_df)

# Model summary
summary(model_2)
```

Call:

```
lm(formula = npp ~ age + biomass + temp + precip + teb + leaf,
    data = forest_df)
```

Residuals:

| Min      | 1Q      | Median | 3Q     | Max     |
|----------|---------|--------|--------|---------|
| -1309.94 | -209.40 | -42.65 | 167.71 | 2898.47 |

Coefficients:

|             | Estimate   | Std. Error | t value | Pr(> t )     |
|-------------|------------|------------|---------|--------------|
| (Intercept) | 7.511e+02  | 3.785e+01  | 19.845  | < 2e-16 ***  |
| age         | -5.004e+00 | 2.829e-01  | -17.690 | < 2e-16 ***  |
| biomass     | 3.180e-02  | 1.228e-03  | 25.905  | < 2e-16 ***  |
| temp        | 2.112e+01  | 2.173e+00  | 9.718   | < 2e-16 ***  |
| precip      | 1.120e-01  | 2.851e-02  | 3.929   | 9.02e-05 *** |
| teb         | -2.255e+00 | 1.092e+00  | -2.066  | 0.039 *      |
| leafneedle  | -2.634e+02 | 2.258e+01  | -11.666 | < 2e-16 ***  |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 354.3 on 1213 degrees of freedom  
 Multiple R-squared: 0.6359, Adjusted R-squared: 0.6341  
 F-statistic: 353.1 on 6 and 1213 DF, p-value: < 2.2e-16

```
# Check VIF again
vif_values2 <- vif(model_2)
vif_values2
```

```
age biomass temp precip teb leaf
1.932806 2.056724 1.696787 1.803266 1.166162 1.234906
```

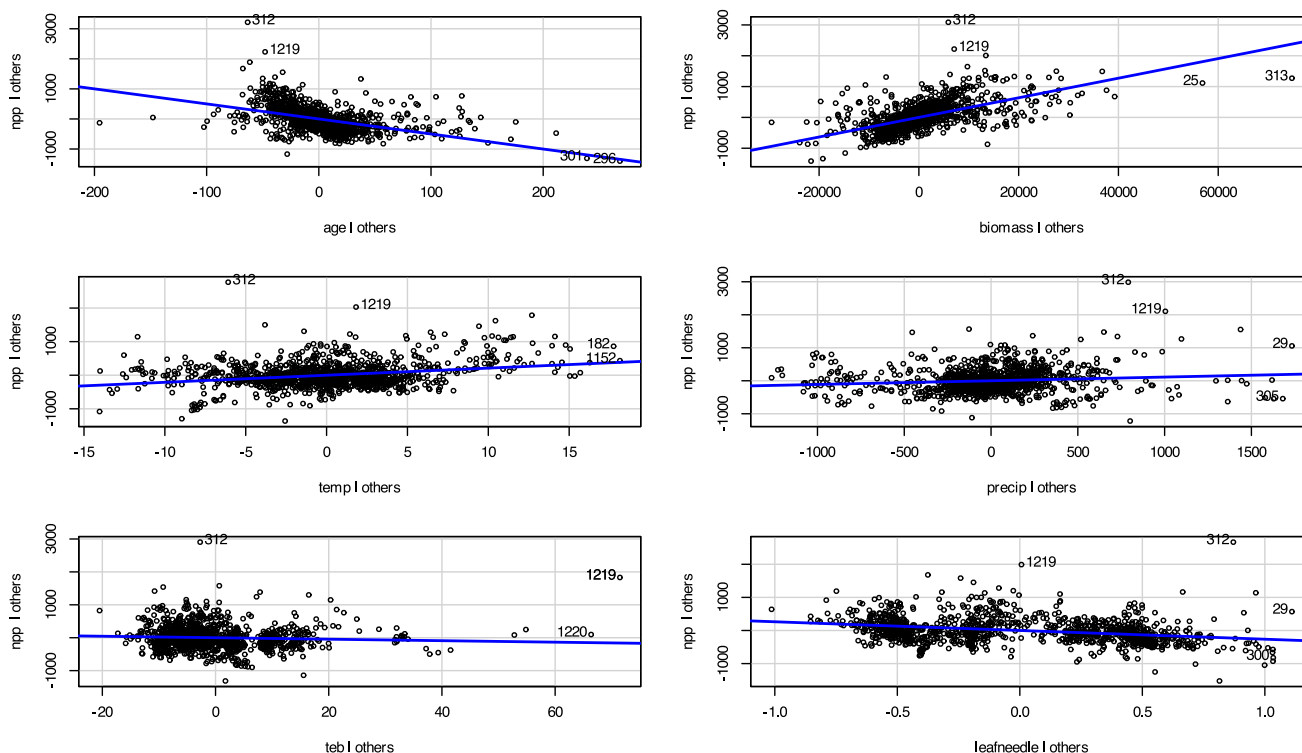
### 3. Exploring Variable Transformations

#### Check the Shape of Relationships with Partial Regression Plots

```
# Create partial regression plots (Added Variable Plots)
# This helps us see the relationship between each predictor and response
# after accounting for other variables

par(mfrow = c(2, 3))
avPlots(model_2, main = "Partial Regression Plots")
```

Partial Regression Plots



```
par(mfrow = c(1, 1))
```

#### Apply Log Transformation to age

The original analysis found that age showed a curvy relationship. Let's try log transformation: Really what you should do is log transform of the response variable first..

```
# Create dataset with log-transformed age
forest_df <- forest_df %>%
  mutate(log_age = log10(age))

# Model 3: With log-transformed age
model_3 <- lm(npp ~ log_age + biomass + temp + precip + teb + leaf,
  data = forest_df)

# Model summary
summary(model_3)
```

```
Call:
lm(formula = npp ~ log_age + biomass + temp + precip + teb +
    leaf, data = forest_df)

Residuals:
    Min       1Q   Median       3Q      Max
-1282.97  -203.31   -23.13   163.75  2810.76

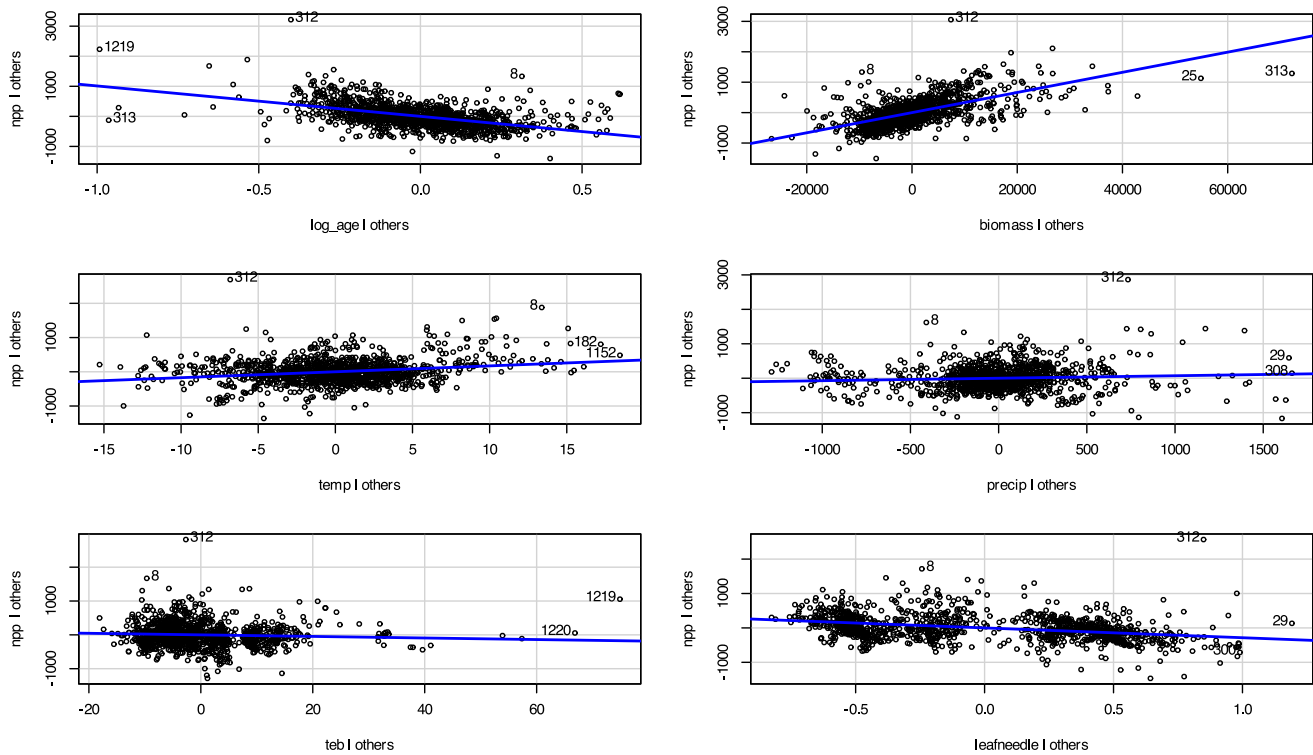
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.237e+03  8.961e+01  24.962 < 2e-16 ***
log_age      -1.012e+03  5.064e+01 -19.973 < 2e-16 ***
biomass       3.313e-02  1.194e-03  27.755 < 2e-16 ***
temp         1.723e+01  2.159e+00   7.982 3.33e-15 ***
precip       7.194e-02  2.781e-02   2.587 0.00981 **
teb          -2.315e+00  1.061e+00  -2.183 0.02925 *
leafneedle   -2.852e+02  2.177e+01 -13.105 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 344.7 on 1213 degrees of freedom
Multiple R-squared:  0.6553,    Adjusted R-squared:  0.6536
F-statistic: 384.4 on 6 and 1213 DF,  p-value: < 2.2e-16
```

```
# Compare partial regression plots
par(mfrow = c(2, 3))
avPlots(model_3, main = "Partial Regression Plots (Log age)")
```



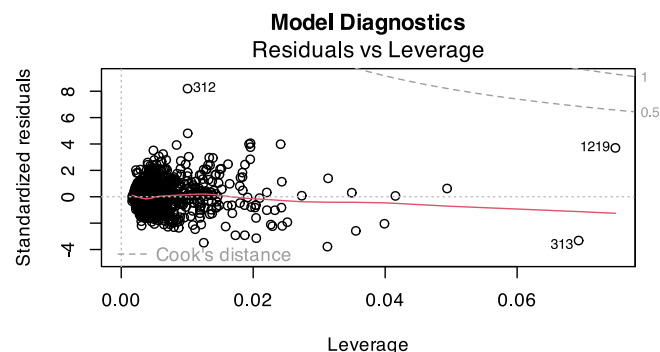
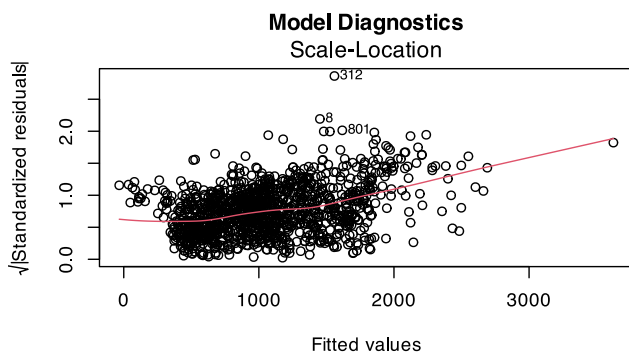
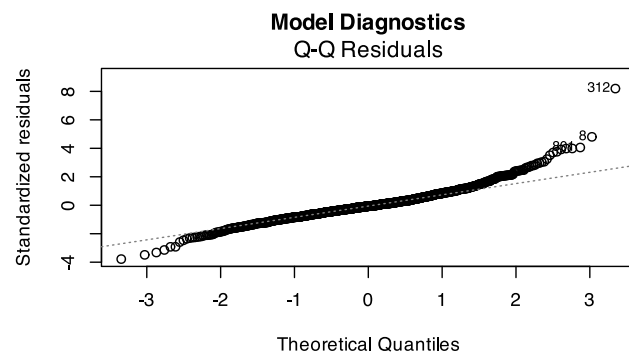
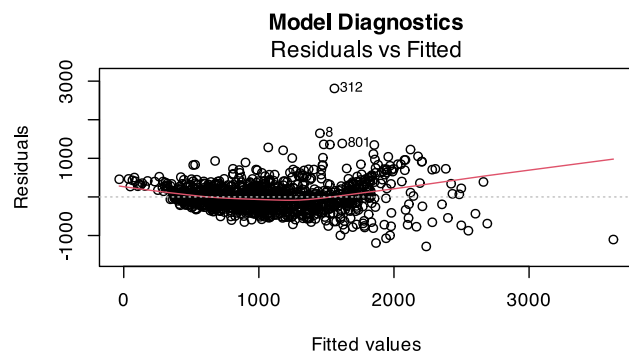
## Partial Regression Plots (Log age)



```
par(mfrow = c(1, 1))
```

## 4. Model Diagnostics and Assumption Checking

```
# Create diagnostic plots
par(mfrow = c(2, 2))
plot(model_3, main = "Model Diagnostics")
```

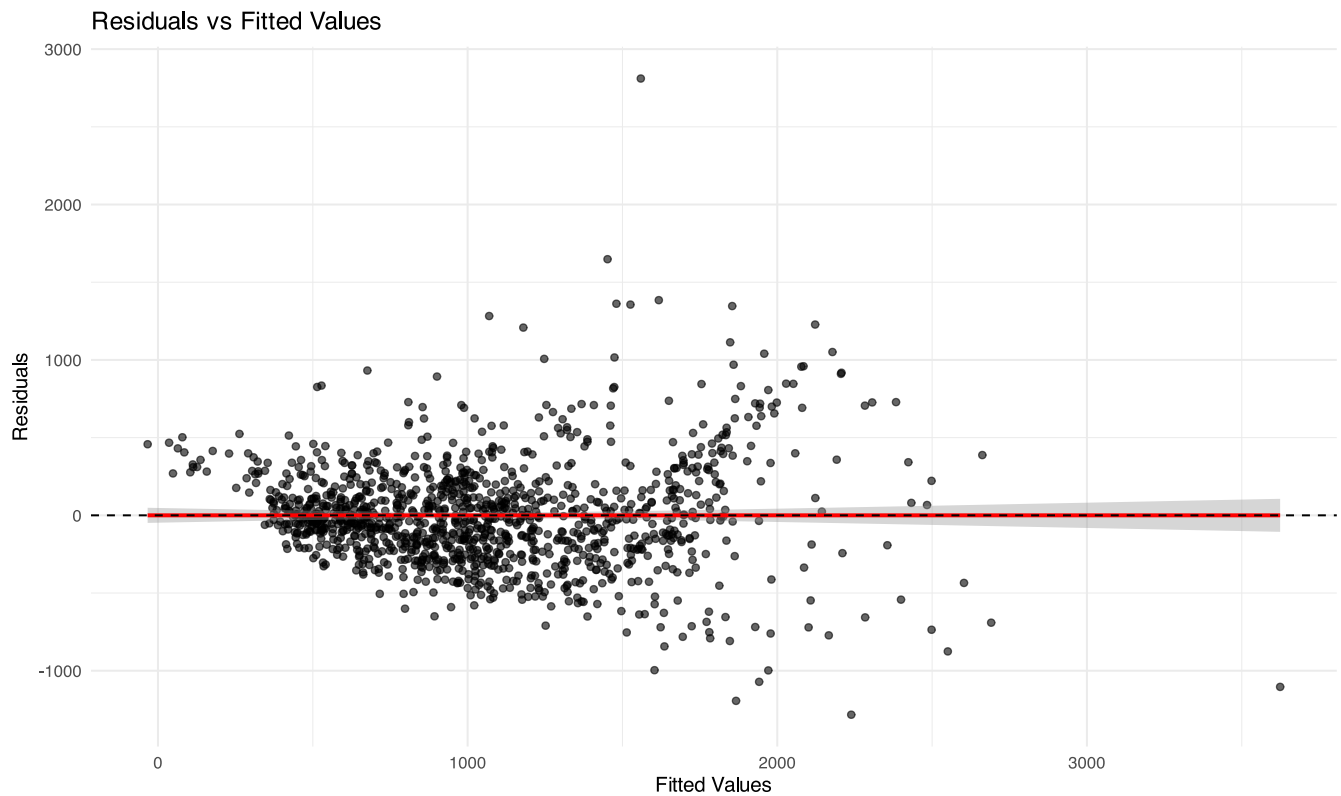


```
par(mfrow = c(1, 1))
```

## Residuals - its the upper left above

```
# Check for normality of residuals
residuals_data <- data.frame(
  Fitted = fitted(model_3),
  Residuals = residuals(model_3),
  Standardized_Residuals = rstandard(model_3)
)

# Residuals vs Fitted plot using ggplot
ggplot(residuals_data, aes(x = Fitted, y = Residuals)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", color = "red") +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(
    title = "Residuals vs Fitted Values",
    x = "Fitted Values",
    y = "Residuals"
  ) +
  theme_minimal()
```



### Try Response Variable Transformation

Following the original analysis, let's try a cube root transformation of npp:

```
# Model 4: Cube root transformation of npp
forest_df <- forest_df %>%
  mutate(npp_cuberoot = npp^(1/3))

model_4 <- lm(npp_cuberoot ~ log_age + biomass + temp + precip +
  teb + leaf, data = forest_df)

summary(model_4)
```

Call:

```
lm(formula = npp_cuberoot ~ log_age + biomass + temp + precip +
  teb + leaf, data = forest_df)
```

Residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -4.0936 | -0.5916 | -0.0030 | 0.6463 | 5.0795 |

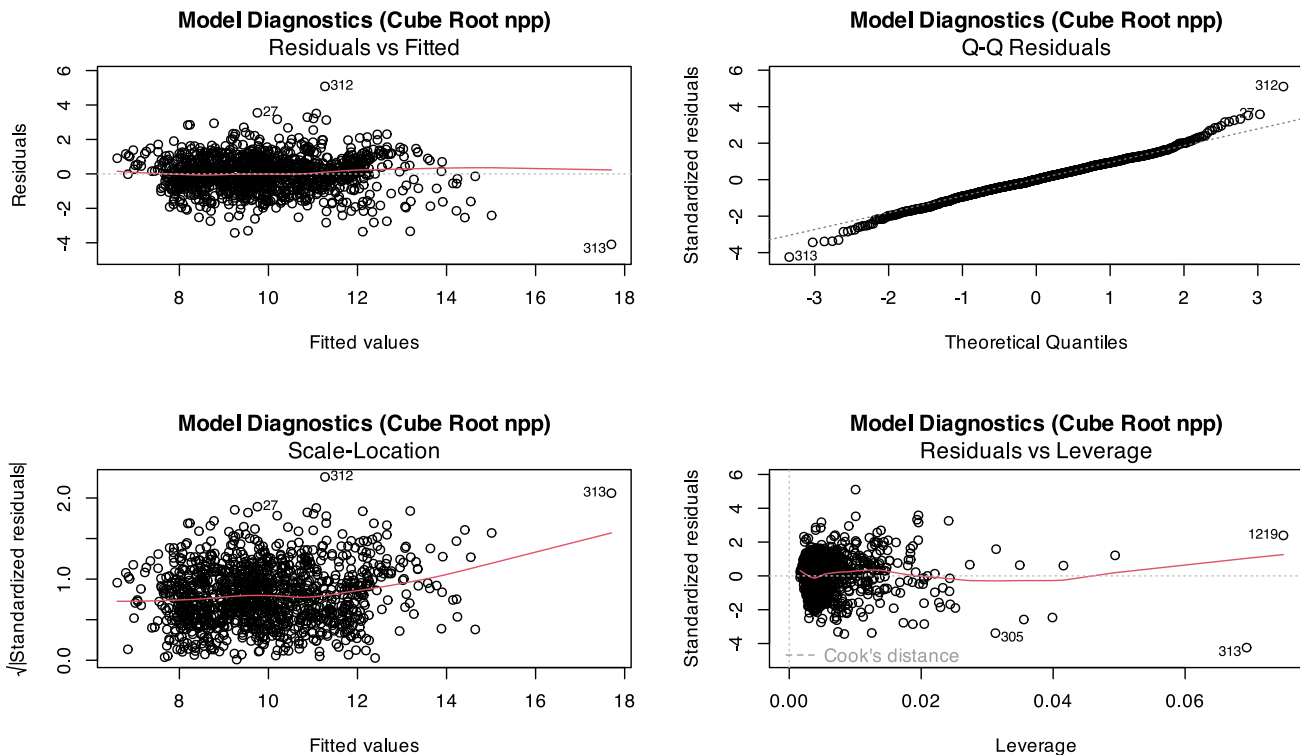
Coefficients:

|             | Estimate   | Std. Error | t value | Pr(> t )     |
|-------------|------------|------------|---------|--------------|
| (Intercept) | 1.391e+01  | 2.601e-01  | 53.489  | < 2e-16 ***  |
| log_age     | -3.151e+00 | 1.470e-01  | -21.436 | < 2e-16 ***  |
| biomass     | 1.032e-04  | 3.464e-06  | 29.791  | < 2e-16 ***  |
| temp        | 4.507e-02  | 6.265e-03  | 7.194   | 1.10e-12 *** |
| precip      | 6.361e-05  | 8.072e-05  | 0.788   | 0.431        |
| teb         | -1.273e-02 | 3.078e-03  | -4.135  | 3.79e-05 *** |
| leafneedle  | -1.029e+00 | 6.317e-02  | -16.283 | < 2e-16 ***  |

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 1 on 1213 degrees of freedom
Multiple R-squared:  0.6793,    Adjusted R-squared:  0.6777
F-statistic: 428.2 on 6 and 1213 DF,  p-value: < 2.2e-16
```

```
# Check diagnostics
par(mfrow = c(2, 2))
plot(model_4, main = "Model Diagnostics (Cube Root npp)")
```



```
par(mfrow = c(1, 1))
```

## 5. Model Simplification and Comparison

### Remove Non-significant Variables

```
# Model 5: Remove non-significant Precipitation
model_5 <- lm(npp_cuberoot ~ log_age + biomass + temp + teb + leaf,
              data = forest_df)

summary(model_5)
```

```
Call:
lm(formula = npp_cuberoot ~ log_age + biomass + temp + teb +
    leaf, data = forest_df)
```

```

Residuals:
    Min       1Q   Median       3Q      Max
-4.1085 -0.5919  0.0019  0.6459  5.1261

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.396e+01  2.532e-01  55.120 < 2e-16 ***
log_age      -3.159e+00  1.465e-01 -21.559 < 2e-16 ***
biomass       1.040e-04  3.306e-06  31.465 < 2e-16 ***
temp         4.719e-02  5.658e-03   8.342 < 2e-16 ***
teb          -1.296e-02  3.064e-03  -4.231 2.5e-05 ***
leafneedle   -1.040e+00  6.150e-02 -16.910 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1 on 1214 degrees of freedom
Multiple R-squared:  0.6791,    Adjusted R-squared:  0.6778
F-statistic: 513.9 on 5 and 1214 DF,  p-value: < 2.2e-16

```

```

# Compare models using AIC
model_comparison <- data.frame(
  Model = c("Model 4 (Full)", "Model 5 (No precip)"),
  AIC = c(AIC(model_4), AIC(model_5)),
  R_squared = c(summary(model_4)$r.squared, summary(model_5)$r.squared),
  Adj_R_squared = c(summary(model_4)$adj.r.squared, summary(model_5)$adj.r.squared)
)

model_comparison

```

|   | Model               | AIC      | R_squared | Adj_R_squared |
|---|---------------------|----------|-----------|---------------|
| 1 | Model 4 (Full)      | 3472.092 | 0.6792946 | 0.6777082     |
| 2 | Model 5 (No precip) | 3470.717 | 0.6791304 | 0.6778089     |

## Model Performance and Interpretation

```

# Get tidy summary of final model
summary(model_5, conf.int = TRUE)

```

```

Call:
lm(formula = npp_cuberooot ~ log_age + biomass + temp + teb +
    leaf, data = forest_df)

```

```

Residuals:
    Min       1Q   Median       3Q      Max
-4.1085 -0.5919  0.0019  0.6459  5.1261

```

```

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.396e+01  2.532e-01  55.120 < 2e-16 ***
log_age      -3.159e+00  1.465e-01 -21.559 < 2e-16 ***
biomass       1.040e-04  3.306e-06  31.465 < 2e-16 ***
temp         4.719e-02  5.658e-03   8.342 < 2e-16 ***
teb          -1.296e-02  3.064e-03  -4.231 2.5e-05 ***

```

```
leafneedle -1.040e+00 6.150e-02 -16.910 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1 on 1214 degrees of freedom
Multiple R-squared:  0.6791,    Adjusted R-squared:  0.6778
F-statistic: 513.9 on 5 and 1214 DF,  p-value: < 2.2e-16
```

## using the sensemaker package

```
library(sensemaker)
# Calculate partial R-squared for all variables at once
partial_r2_sensemaker <- partial_r2(model_5)
partial_r2_sensemaker
```

```
(Intercept)    log_age    biomass    temp    teb    leafneedle
0.71450239 0.27686109 0.44919401 0.05421045 0.01453129 0.19064332
```

```
# Calculate partial R-squared for each variable
# Using the car package
print("Using car package Anova() with Type III sums of squares:")
```

```
[1] "Using car package Anova() with Type III sums of squares:"
```

```
anova_type2 <- Anova(model_5, type = "III")
print(anova_type2)
```

Anova Table (Type III tests)

```
Response: npp_cuberoot
      Sum Sq   Df F value    Pr(>F)
(Intercept) 3039.52    1 3038.225 < 2.2e-16 ***
log_age      464.99    1  464.792 < 2.2e-16 ***
biomass      990.47    1  990.043 < 2.2e-16 ***
temp         69.61    1   69.584 < 2.2e-16 ***
teb          17.91    1   17.901 2.502e-05 ***
leaf        286.08    1  285.957 < 2.2e-16 ***
Residuals   1214.52 1214
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Convert F-statistics to partial R-squared
# Partial R² = F * df_num / (F * df_num + df_den)
f_stats <- anova_type2$`F value`[!is.na(anova_type2$`F value`)]
df_num <- anova_type2$Df[!is.na(anova_type2$`F value`)]
df_den <- anova_type2$Df[nrow(anova_type2)] # Residual df

partial_r2_from_f <- f_stats * df_num / (f_stats * df_num + df_den)

results_table <- data.frame(
  Variable = rownames(anova_type2)[!is.na(anova_type2$`F value`)],
```

```

F_statistic = f_stats,
p_value = anova_type2$`Pr(>F)`[!is.na(anova_type2$`F value`)],
Partial_R_squared = partial_r2_from_f
)

print("Complete results with partial R-squared:")

```

```
[1] "Complete results with partial R-squared:"
```

```
print(results_table)
```

|   | Variable    | F_statistic | p_value       | Partial_R_squared |
|---|-------------|-------------|---------------|-------------------|
| 1 | (Intercept) | 3038.22472  | 0.000000e+00  | 0.71450239        |
| 2 | log_age     | 464.79225   | 1.532052e-87  | 0.27686109        |
| 3 | biomass     | 990.04285   | 2.085787e-159 | 0.44919401        |
| 4 | temp        | 69.58365    | 1.967636e-16  | 0.05421045        |
| 5 | teb         | 17.90111    | 2.501886e-05  | 0.01453129        |
| 6 | leaf        | 285.95672   | 9.100842e-58  | 0.19064332        |

## 6. Alternative Approach: Standardized Variables

Following the original analysis, let's also try the standardized approach:

```

# Create standardized variables
forest_standardized <- forest_df %>%
  mutate(
    npp_sqrt_scaled = scale(sqrt(npp))[,1],
    log_age_scaled = scale(log10(age))[,1],
    biomass_scaled = scale(biomass)[,1],
    temp_scaled = scale(temp)[,1],
    precip_scaled = scale(precip)[,1],
    teb_scaled = scale(teb)[,1]
  )

# Standardized model
model_std <- lm(npp_sqrt_scaled ~ log_age_scaled + biomass_scaled +
  temp_scaled * precip_scaled + teb_scaled,
  data = forest_standardized)

# Summary
summary(model_std)

```

Call:

```
lm(formula = npp_sqrt_scaled ~ log_age_scaled + biomass_scaled +
  temp_scaled * precip_scaled + teb_scaled, data = forest_standardized)
```

Residuals:

|  | Min      | 1Q       | Median   | 3Q      | Max     |
|--|----------|----------|----------|---------|---------|
|  | -3.14438 | -0.38836 | -0.03068 | 0.39178 | 3.01216 |

Coefficients:

|  | Estimate | Std. Error | t value | Pr(> t ) |
|--|----------|------------|---------|----------|
|--|----------|------------|---------|----------|

```

(Intercept)          -0.064952    0.020075   -3.235 0.001247 **
log_age_scaled       -0.543699    0.024500  -22.192 < 2e-16 ***
biomass_scaled        0.691415    0.025313   27.315 < 2e-16 ***
temp_scaled          0.215693    0.023304    9.256 < 2e-16 ***
precip_scaled        -0.008188    0.028714   -0.285 0.775563
teb_scaled           -0.071775    0.018879   -3.802 0.000151 ***
temp_scaled:precip_scaled 0.112727    0.017068    6.605 5.94e-11 ***

```

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6113 on 1213 degrees of freedom  
Multiple R-squared: 0.6281, Adjusted R-squared: 0.6263  
F-statistic: 341.5 on 6 and 1213 DF, p-value: < 2.2e-16

## 7. Key Findings and Conclusions

### 1. MULTICOLLINEARITY:

- Growing season length and temperature were highly correlated
- Removed growing season to address multicollinearity

### 2. VARIABLE TRANSFORMATIONS:

- Log transformation of age improved model fit
- Cube root transformation of npp addressed assumption violations

### 3. FINAL MODEL RESULTS:

- Significant predictors: age (negative), biomass (positive), temp (positive)
- teb had negative effect, Leaf type differences were significant

### 4. BIOLOGICAL INTERPRETATION:

- Younger stands had higher npp (for given biomass)
- Higher biomass associated with higher npp
- temp positively related to npp
- Coniferous forests had lower npp than broadleaf forests

## References and Additional Notes

This analysis is based on:

- **Michaletz, S.T., Cheng, D., Kerkhoff, A.J. & Enquist, B.J.** (2014). Convergence of terrestrial plant production across global climate gradients. *Nature*, 512, 39-43.

### Key Learning Points:

1. **Multicollinearity Detection:** Use VIF values and correlation matrices
2. **Variable Transformations:** Log and power transformations can improve model fit
3. **Model Diagnostics:** Always check residual plots and assumption violations
4. **Model Comparison:** Use AIC and other criteria for model selection
5. **Interpretation:** Focus on biologically meaningful relationships

```

# Session information for reproducibility
sessionInfo()

```

R version 4.5.1 (2025-06-13)  
Platform: aarch64-apple-darwin20



Running under: macOS Sequoia 15.6

Matrix products: default

BLAS: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib

LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;

LAPACK version 3.12.1

locale:

[1] en\_US.UTF-8/en\_US.UTF-8/en\_US.UTF-8/C/en\_US.UTF-8/en\_US.UTF-8

time zone: America/Chicago

tzcode source: internal

attached base packages:

[1] stats graphics grDevices utils datasets methods base

other attached packages:

|                     |                 |                    |               |
|---------------------|-----------------|--------------------|---------------|
| [1] sensemakr_0.1.6 | see_0.11.0      | performance_0.15.0 | broom_1.0.9   |
| [5] GGally_2.3.0    | corrplot_0.95   | car_3.1-3          | carData_3.0-5 |
| [9] lubridate_1.9.4 | forcats_1.0.0   | stringr_1.5.1      | dplyr_1.1.4   |
| [13] purrr_1.1.0    | readr_2.1.5     | tidyr_1.3.1        | tibble_3.3.0  |
| [17] ggplot2_3.5.2  | tidyverse_2.0.0 |                    |               |

loaded via a namespace (and not attached):

|                        |                   |                |                    |
|------------------------|-------------------|----------------|--------------------|
| [1] gtable_0.3.6       | xfun_0.52         | insight_1.3.1  | lattice_0.22-7     |
| [5] tzdb_0.5.0         | vctrs_0.6.5       | tools_4.5.1    | generics_0.1.4     |
| [9] parallel_4.5.1     | pkgconfig_2.0.3   | Matrix_1.7-3   | RColorBrewer_1.1-3 |
| [13] S7_0.2.0          | lifecycle_1.0.4   | compiler_4.5.1 | farver_2.1.2       |
| [17] codetools_0.2-20  | htmltools_0.5.8.1 | yaml_2.3.10    | Formula_1.2-5      |
| [21] pillar_1.11.0     | crayon_1.5.3      | abind_1.4-8    | nlme_3.1-168       |
| [25] ggstats_0.10.0    | tidyselect_1.2.1  | digest_0.6.37  | stringi_1.8.7      |
| [29] labeling_0.4.3    | splines_4.5.1     | fastmap_1.2.0  | grid_4.5.1         |
| [33] cli_3.6.5         | magrittr_2.0.3    | utf8_1.2.6     | withr_3.0.2        |
| [37] scales_1.4.0      | backports_1.5.0   | bit64_4.6.0-1  | timechange_0.3.0   |
| [41] rmarkdown_2.29    | bit_4.6.0         | hms_1.1.3      | evaluate_1.0.4     |
| [45] knitr_1.50        | mgcv_1.9-3        | rlang_1.1.6    | glue_1.8.0         |
| [49] rstudioapi_0.17.1 | vroom_1.6.5       | jsonlite_2.0.0 | R6_2.6.1           |